

Project Executable Files

Date	1 July 2026
Team ID	SWTID-2026-4993
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Project Executable Files

This section provides a detailed record of all executable and deployable components developed for the OptiCrop project. Each file has been systematically organized, properly named, and uploaded to GitHub for easy access. The evaluator can review and test the live deployed version of the application using the provided repository link.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / dependency file	Yes
4	Database schema / seed files (if applicable)	NA — CSV dataset used
5	Environment configuration file (.env.example or similar)	NA — No API keys required
6	Deployed application URL (if hosted)	Yes — Deployed on Render
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA — Web application
8	Dockerfile / Containerization config (if applicable)	NA — Render handles deployment

9	Test files and test results	Yes — Performance Testing PDF
10	Demo video or walkthrough (if required)	Yes — Demo video uploaded to Drive



Step 2: File / Folder Structure

```
AI-ML-and-GEN-AI-Track-Project-Template/ ■■■ 1.Brainstorming & Ideation/ ■■■■ Brainstorming &
Idea Prioritization.pdf ■■■■ Define Problem Statements.pdf ■■■■ Empathy Map.pdf ■■■■ ER
Diagram.pdf ■■■ 2.Requirement Analysis/ ■■■■ Customer Journey Map.pdf ■■■■ Data Flow
Diagram.pdf ■■■■ Solution Requirements.pdf ■■■■ Technology Stack.pdf ■■■ 3.Project Design
Phase/ ■■■■ Problem-Solution Fit.pdf ■■■■ Proposed Solution.pdf ■■■■ Solution
Architecture.pdf ■■■ 4.Project Planning Phase/ ■■■■ Project Planning.pdf ■■■ 5.Project
Development Phase/ ■■■■ OptiCrop.ipynb ■■■■ app.py ■■■■ model.pkl ■■■■
Crop_recommendation.csv ■■■■ requirements.txt ■■■■ Procfile ■■■■ Coding & Solution.pdf ■■■■
templates/ ■■■■ home.html ■■■■ about.html ■■■■ findyourcrop.html ■■■ 6.Project Testing/ ■■■■
Performance Testing.pdf ■■■ 7.Project Documentation/ ■■■■ Project Executable Files.pdf ■■■■
Sample Project Documentation.pdf ■■■ 8.Project Demonstration/ ■■■■ Communication.pdf ■■■■
Demonstration of Proposed Features.pdf ■■■■ Project Demo Planning.pdf ■■■■ Scalability & Future
Plan.pdf ■■■■ Team Involvement in Demonstration.pdf ■■■ README.md
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://opticrop-project-b8ec.onrender.com
Platform / Hosting Provider	Render
Repository Link	https://github.com/Yaddala-Geethanjali/OptiCrop-Project
Demo Video Link	https://drive.google.com/file/d/1S6g21957saldqCXZ9qprIFS_h5Km63XH/view?usp=sharing

Step 4: Run Instructions

Step 1: Clone the repository `git clone < https://github.com/Yaddala-Geethanjali/OptiCrop-Project>`

Step 2 : Open the OptiCrop project folder in VS Code.

Step 3: Install dependencies:

```
pip install -r requirements.txt
```

Step 4: Run the application:

```
python app.py
```

Step 5: Open a browser and visit:

<http://127.0.0.1:5000>

Step 6: Enter the required values and click Predict to view the recommended crop.



Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Initial loading of the application may take a little longer on the free hosting service.	Wait for about 30–60 seconds during the first launch. After that, the application responds fast
2	The Logistic Regression model may display a convergence warning when the iteration limit is too low.	The issue was resolved by increasing the <code>max_iter</code> value during model training.
3	User login and authentication features are not included in the current version.	These features are planned for future updates and are outside the scope of the present project.